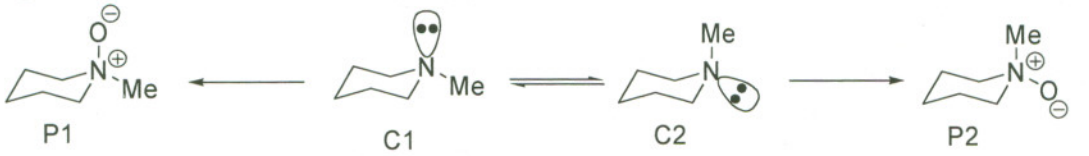
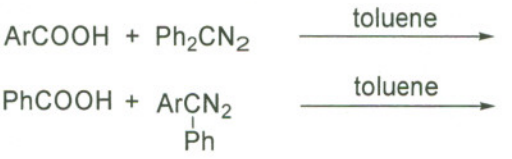
Organic Cume, October 8, 2004

1. The below oxidation reaction gives a product ratio P1:P2 of 19:1. However, Keq=[C2]/[C1]=0.095. Show how the Curtin-Hammett principle explains the overall dichotomy between the equilibrium constant of the conformation of reactants and the product ratio. You should draw an energy diagram of this process shown below and exactly label all relevant components. Clearly indicate why, based on the diagram what determines the product ratio.

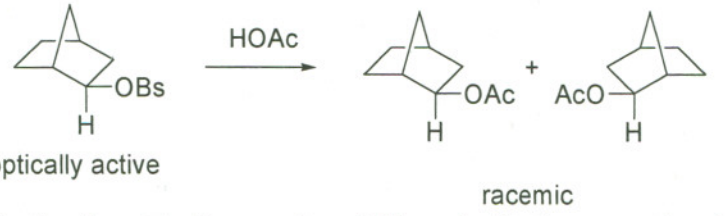


2. Indicate whether Hammett plots for rates of each of the following reactions (i.e. Ar is the substituted benzene) should have positive or negative ρ. Estimate the magnitude of the slopes ($>|1|$ or $<|1|$). Decide whether the best fit to straight lines would be obtained by using values of σ, $\sigma+$ or $\sigma-$.



3. Give a More O'Farrell-Jencks Diagram for substitution reactions.

4. Give the mechanism of the following reaction. Be sure to justify the observed stereochemistry in the product. Also explain why the endo reactant (not shown) is 350 times slower than the exo reactant shown below.



optically active

racemic

5. Show the effect of solvent in the nucleophilic substitution reaction between iodide and chloromethane by drawing two potential energy diagrams using methanol as one solvent vs. acetone as a second solvent.